

$^{34}\text{S}(\text{n},\text{n}),(\text{n},\gamma)$:resonances [1984Ca14,2018MuZY](#)

2018MuZY: Evaluation of neutron resonance energies, J^π values, width parameters, and resonance strengths for nuclei of $Z=1$ -60.
1984Ca14: 30-1500-keV neutrons were produced from the Oak Ridge Electron Linear Accelerator (ORELA). The target was 94.33%-enriched ^{34}S with a 5.54% impurity of ^{32}S . Neutrons were detected using a 2-cm-thick and 7.5-cm in diameter NE110 proton-recoil detector. Prompt γ rays from neutron capture were detected using two nonhydrogenous fluorocarbon scintillators (NE-226). Measured $\sigma(E_n)$. Deduced 81 resonances, J , π , L , widths and spectroscopic factors by comparing the R-matrix predictions of the cross section with the data.

 ^{35}S Levels

All resonance parameters including resonance neutron energies, J , L , $g\Gamma_n$, Γ_γ , and resonance strengths are from the compilation and evaluation in [2018MuZY](#), unless otherwise noted. $g\Gamma_n=(2J+1)\Gamma_n/2$.

$E(\text{level})^\dagger$	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	$E_n(\text{lab})$ (keV)	Comments
6985.54? 4	1/2		0	-0.305	$E(\text{level})$: fictitious level from analysis of resonance spectrum below the $S(\text{n})$ threshold. $\Gamma_\gamma=0.38$ eV.
7018.89 4	1/2	0.025 eV 1	[2]	34.03 1	
7072.66 5		0.14 eV 4		89.40 3	
7074.33 6		0.084 eV 3		91.12 4	
7097.72 30		0.009 eV 3	[2]	115.2 3	
7099.85 20		0.033 eV 5	[2]	117.4 2	
7100.73 4	3/2	0.70 eV 4	[1]	118.30 1	$g\Gamma_n=372$ eV 3; $\Gamma_\gamma=0.35$ eV 2.
7142.39 11		0.420 eV 10	[2]	161.2 1	$E_n(\text{lab})$ (keV): 161.2 1 in 1984Ca14 . $g\Gamma_n<1$ eV.
7210.42 5		0.300 eV 20	[1]	231.25 2	$g\Gamma_n=62$ eV 4.
7218.0 4		0.330 eV 20	[2]	239.0 4	$g\Gamma_n<4$ eV.
7234.0 5		0.190 eV 10	[2]	255.5 5	$g\Gamma_n<4$ eV.
7239.8 6		0.060 eV 20	[2]	261.5 6	$E_n(\text{lab})$ (keV): Revised from a duplicate 239.0 6 in 2018MuZY by evaluators based on the original data in 1984Ca14 . $g\Gamma_n<2$ eV.
7253.2 6		0.350 eV 20	[2]	275.3 6	$g\Gamma_n<1$ eV.
7275.93 5	1/2	2.8 eV 7	0	298.70 3	$g\Gamma_n=9600$ eV 150; $\Gamma_\gamma=2.82$ eV 67.
7279.13 4		0.100 eV 30		302	
7289.8 6		0.310 eV 30	[2]	313.0 6	$E_n(\text{lab})$ (keV): 313.0 8 in 1984Ca14 . $g\Gamma_n<6$ eV.
7294.19 6	3/2	0.46 eV 16	1	317.51 5	$g\Gamma_n=2500$ eV 25; $\Gamma_\gamma=0.23$ eV 8.
7331.00 5	1/2	0.30 eV 30	0	355.41 2	$g\Gamma_n=4810$ eV 80; $\Gamma_\gamma=0.3$ eV 3.
7338.18 20		1.06 eV 7	[1]	362.8 2	$g\Gamma_n=230$ eV 10.
7343.91 20		0.720 eV 30	[2]	368.7 2	$g\Gamma_n<2$ eV.
7347.2 5		0.44 eV 6	[2]	372.1 5	$g\Gamma_n<2$ eV.
7347.9 5		0.30 eV 5	[2]	372.8 5	$g\Gamma_n<2$ eV.
7353.9 5		0.44 eV 4	[2]	379.0 5	$g\Gamma_n<10$ eV.
7357.6 5		0.16 eV 4	[2]	382.8 5	$g\Gamma_n<6$ eV.
7370.52 6	1/2	3.08 eV 20	1	396.10 4	$E_n(\text{lab})$ (keV): 396.14 4 in 1984Ca14 . $g\Gamma_n=6430$ eV 150; $\Gamma_\gamma=3.08$ eV 20.
7371.97 4		0.120 eV		397.6	
7396.0 6		0.44 eV 33	[2]	422.3 6	
7404.6 6		0.21 eV 5	[2]	431.2 6	$g\Gamma_n<1$ eV.
7408.68 11	3/2	1.820 eV 16	1	435.4 1	$g\Gamma_n=1570$ eV 40; $\Gamma_\gamma=0.91$ eV 8.
7411.55 6			[2]	438.35 5	$g\Gamma_n=4.9$ eV 50.
7416.50 4		0.51 eV 5	[2]	443.45 1	$g\Gamma_n=80$ eV 8.
7429.1 6		0.120 eV 30	[2]	456.4 6	$g\Gamma_n<6$ eV.
7433.56 5		0.035 eV 20	[2]	461.01 2	$g\Gamma_n=50$ eV 5.
7437 4		0.280 eV 30	[1]	464 4	$E_n(\text{lab})$ (keV): 463.9 4 in 1984Ca14 . $g\Gamma_n=296$ eV 30.

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$^{34}\text{S}(\text{n},\text{n}),(\text{n},\gamma)$:resonances **1984Ca14,2018MuZY** (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	$E_n(\text{lab})$ (keV)	Comments
7442.14 5	1/2	0.37 eV 8	0	469.85 2	$g\Gamma_n=1050$ eV 35; $\Gamma_\gamma=0.37$ eV 8.
7462.2 6		0.33 eV 4	[2]	490.5 6	$g\Gamma_n=50$ eV 20.
7481.33 6	1/2	2.1 eV 10	1	510.20 4	$E_n(\text{lab})$ (keV): 510.02 4 in 1984Ca14 .
7494.5 6		0.17 eV 4	[2]	523.8 6	$g\Gamma_n=375$ eV 25; $\Gamma_\gamma=2.1$ eV 1.
7542.3 6		0.210 eV 30	[2]	573.0 6	$g\Gamma_n<4$ eV.
7604.0 6		0.50 eV 8	[2]	636.5 6	$g\Gamma_n<1$ eV.
7609.26 5	3/2	1.02 eV 12	1	641.93 3	$g\Gamma_n<10$ eV.
7648.9 6		0.210 eV 30	[2]	682.7 6	$g\Gamma_n=2520$ eV 60; $\Gamma_\gamma=0.51$ eV 6.
7655.7 6		0.80 eV 5	[2]	689.7 6	$g\Gamma_n<30$ eV.
7663.91 11		0.80 eV 8	[2]	698.2 1	$g\Gamma_n<10$ eV.
7678.3 7		0.45 eV 7	[2]	713.0 7	$g\Gamma_n=305$ eV 50; $\Gamma_\gamma=0.80$ eV 8.
7731.0 6		0.31 eV 13	[2]	767.3 6	$g\Gamma_n<4$ eV.
7749.7 4	(3/2)	2.65 eV 18	[1]	786.5 4	$E_n(\text{lab})$ (keV): 767.3 7 in 1984Ca14 .
7761.5 6	3/2		1	798.7 6	$g\Gamma_n<4$ eV.
7776.17 11	1/2		1	813.8 1	$g\Gamma_n=18$ eV 6; $\Gamma_\gamma=1.33$ eV 9.
7797.99 9	1/2		0	836.27 8	$E_n(\text{lab})$ (keV): 798.65 6 in 1984Ca14 .
7812.24 7	3/2	0.24 eV 19	2	850.94 6	$g\Gamma_n=2535$ eV 300; $\Gamma_\gamma=1.52$ eV 50.
7853.25 7	3/2	0.38 eV 16	1	893.17 6	$g\Gamma_n=860$ eV 60; $\Gamma_\gamma=0.19$ eV 13.
7887.8 2		1.13 eV 19	[2]	902.0 12	$g\Gamma_n=3600$ eV 160; $\Gamma_\gamma=0.39$ eV 18.
7880.8 12		0.67 eV 17	[2]	921.5 12	$g\Gamma_n=2650$ eV 110; $\Gamma_\gamma=0.24$ eV 19.
7889.0 6		0.57 eV 25	[2]	930.0 6	$g\Gamma_n=4950$ eV 160; $\Gamma_\gamma=0.38$ eV 16.
7894.63 7	3/2	0.348 eV 23	1	935.78 6	$g\Gamma_n<3$ eV.
7899.73 7			[2]	941.03 6	$g\Gamma_n<20$ eV.
7934.1 15		0.50 eV 30	[2]	976.4 15	$E_n(\text{lab})$ (keV): 930.0 12 in 1984Ca14 .
7939.5 6		0.89 eV 4	[2]	982.0 6	$g\Gamma_n<20$ eV.
7954.96 7	3/2	3.51 eV 19	1	997.90 6	$g\Gamma_n=4100$ eV 160; $\Gamma_\gamma=0.348$ eV 23.
6993 6	3/2	1.22 eV 18	1	1017.7 6	$E_n(\text{lab})$ (keV): 941.0 3 in 1984Ca14 .
6990 6			[2]	1064.4 6	$g\Gamma_n=176$ eV 16.
8076.46 20			[2]	1123.0 2	$g\Gamma_n<2$ eV.
8077.91 20			[2]	1124.5 2	$E_n(\text{lab})$ (keV): 982.0 15 in 1984Ca14 .
8093.0 5	1/2		1	1140.0 5	$g\Gamma_n<8$ eV.
8143.47 20		0.29 eV 5	[2]	1192.0 2	$E_n(\text{lab})$ (keV): 997.9 1 in 1984Ca14 .
8187.17 30	1/2	5.0 eV 9	1	1237.0 3	$g\Gamma_n=4100$ eV 160; $\Gamma_\gamma=3.51$ eV 19.
8211.25 30		1.61 eV 14	[1]	1261.8 3	$E_n(\text{lab})$ (keV): 1017.7 1 in 1984Ca14 .
8224.2 4	1/2	3.55 eV 30	1	1275.1 4	$E_n(\text{lab})$ (keV): 1064.4 2 in 1984Ca14 .
8228.3 4			[2]	1279.4 4	$g\Gamma_n=2650$ eV 160.
8243.98 30		0.33 eV 4	[2]	1295.5 3	$g\Gamma_n=5480$ eV 390.
8262.4 4	5/2	0.58 eV 23	2	1314.5 4	$E_n(\text{lab})$ (keV): 1124.5 3 in 1984Ca14 .
8273.3 4	1/2	2.58 eV 23	1	1325.7 4	$g\Gamma_n=4730$ eV 320.
8298.27 30			[2]	1351.4 3	$g\Gamma_n=34880$ eV 830.
8333.8 4	3/2	4.4 eV 6	1	1388.0 4	$E_n(\text{lab})$ (keV): 1192.2 2 in 1984Ca14 .
8335.9 4			[2]	1390.1 4	$g\Gamma_n=5510$ eV 290.
8391.3 4	3/2		1	1447.2 4	$g\Gamma_n=2945$ eV 230.

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$^{34}\text{S}(\text{n,n}),(\text{n},\gamma)$:resonances **1984Ca14,2018MuZY (continued)** ^{35}S Levels (continued)

<u>E(level)[†]</u>	<u>$g\Gamma_n\Gamma_\gamma/\Gamma$</u>	<u>L</u>	<u>$E_n(\text{lab})$ (keV)</u>	<u>Comments</u>
8393.6 4	1.02 eV 12	[2]	1449.6 4	$g\Gamma_n=30250$ eV 260.
8406.55 30	0.89 eV 4	[2]	1462.9 3	$g\Gamma_n=88560$ eV 345.
8418.79 30	0.34 eV 8	[2]	1475.5 3	$g\Gamma_n=37960$ eV 340.

[†] From $E_{\text{c.m.}}+S(\text{n})$ where $S(\text{n})=6985.84$ 4 (2021Wa16) and $E_{\text{c.m.}}=E_n(\text{lab})\times m(^{34}\text{S})/[m_n+m(^{34}\text{S})]$.